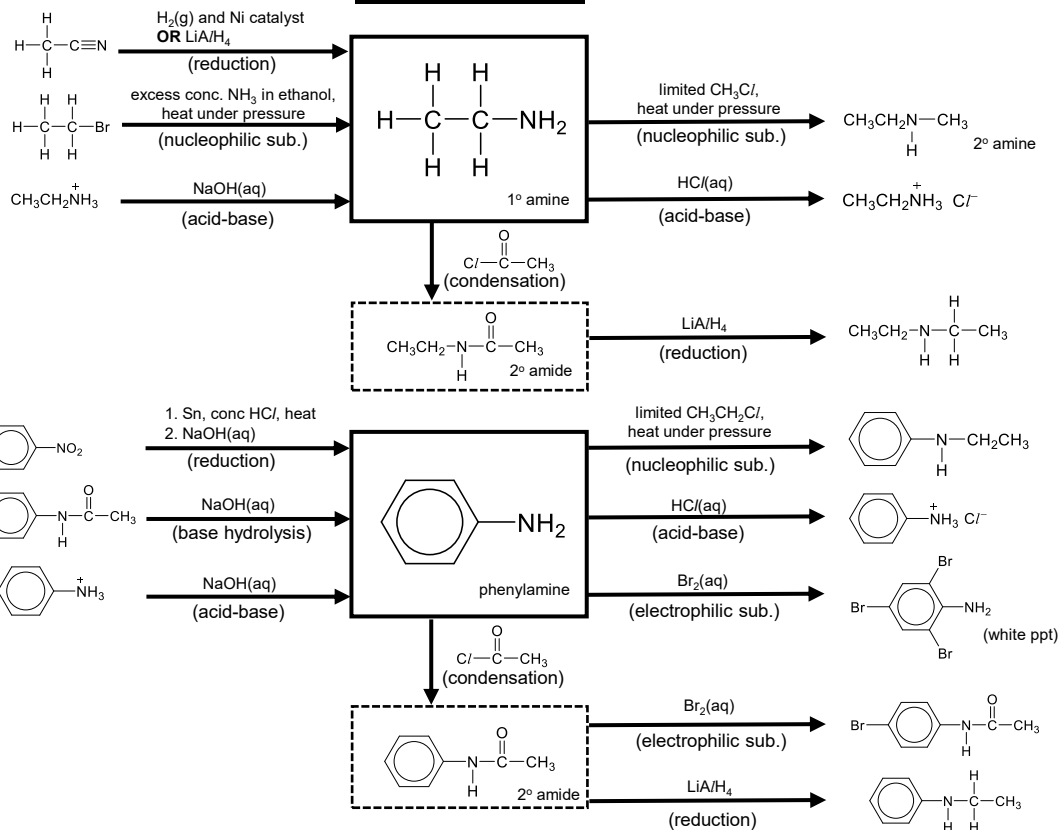


Reactions



Amino acids

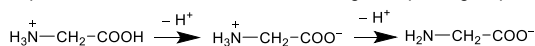
Amino acid exists in an electrically neutral form when in solid state

e.g. $\text{H}_3\text{N}^+-\text{CH}_2-\text{COO}^-$

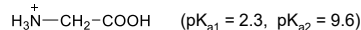
Describe the physical properties of $\text{H}_3\text{N}^+-\text{CH}_2-\text{COO}^-$

melting point	high, due _____, which require large amount of energy to overcome
solubility in water	high, due to formation of _____ with water molecules

Aqueous amino acids can be neutral or charged, depending on pH



Sketch the titration curve when NaOH(aq) is added to



Relative basicity

Label the following from least to most basic

CH_3NH_2	NH_3		$\text{CH}_3-\text{C}(=\text{O})-\text{NH}_2$

Explanation:

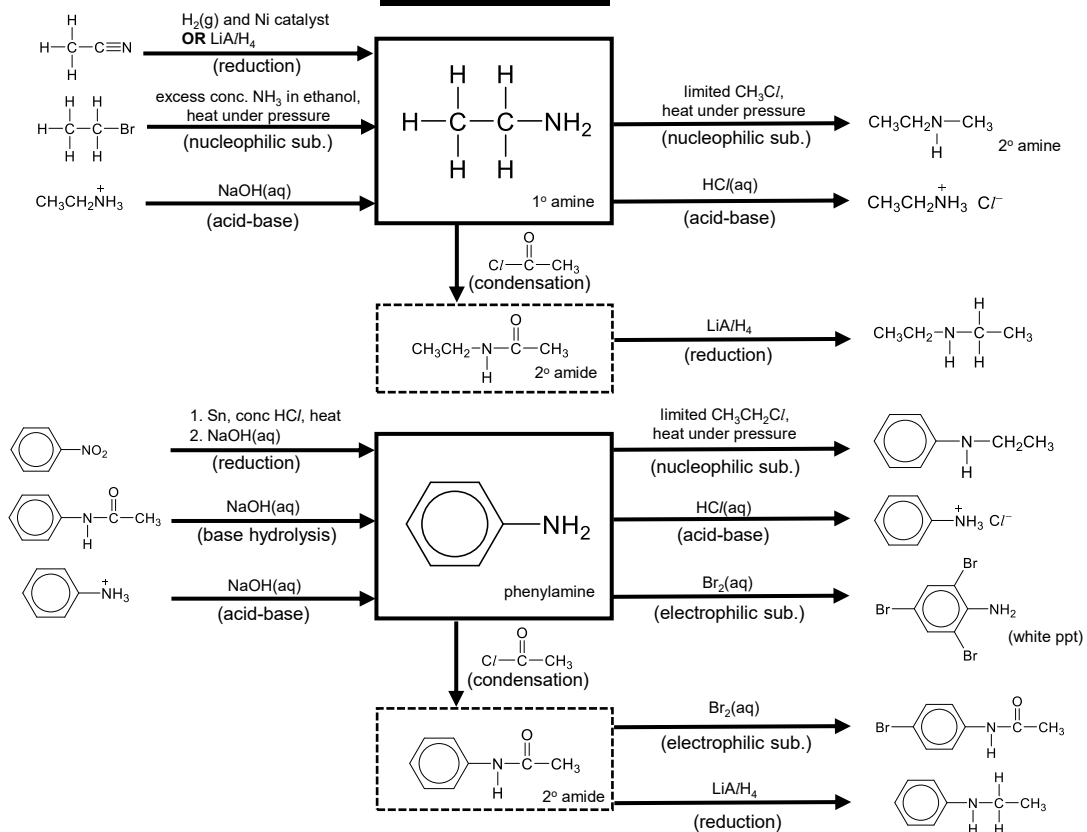
The greater the _____ of lone pair of electrons on N to coordinate with a proton, the greater the basicity.

CH_3NH_2		$\text{CH}_3-\text{C}(=\text{O})-\text{NH}_2$
•	• orbital of N atom containing lone pair of electrons overlaps with π electron cloud of benzene ring	• orbital of N atom containing lone pair of electrons overlaps with π electron cloud of C=O bond
• increases the electron density on N atom	•	•
• hence lone pair of electrons on N more available for coordination with proton	• hence lone pair of electrons on N less available for coordination with proton	• hence lone pair of electrons on N is not available for coordination with proton

Test

Compound	Reagents / Cond ^a	Observation/ explanation
CH_3NH_2 amine with 5C or less	NaOH(aq), warm/ heat	Observation: pungent gas evolved turned damp red litmus paper blue Explanation: amine is volatile, vapourises upon warming (Note: NaOH is not actually required)
$\text{CH}_3(\text{CH}_2)_5\text{NH}_2$ amine with 6C or more	NaOH(aq), warm/ heat	Observation: no pungent gas evolved Explanation: amine is not volatile, does not vaporise upon warming
$\text{CH}_3-\text{C}(=\text{O})-\text{NH}_2$ Amide	NaOH(aq), <u>heat</u>	Observation: pungent gas evolved turned damp red litmus paper blue Explanation:
 Amide	NaOH(aq), <u>heat</u>	Observation: Explanation:
$\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_3^+$ ammonium salt	NaOH(aq), warm/ heat	Observation: pungent gas evolved turned damp red litmus paper blue Explanation:

Reactions



Amino acids

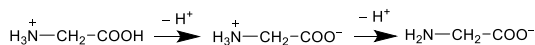
Amino acid exists in an electrically neutral form when in solid state

e.g. $\text{H}_3\text{N}^+-\text{CH}_2-\text{COO}^-$

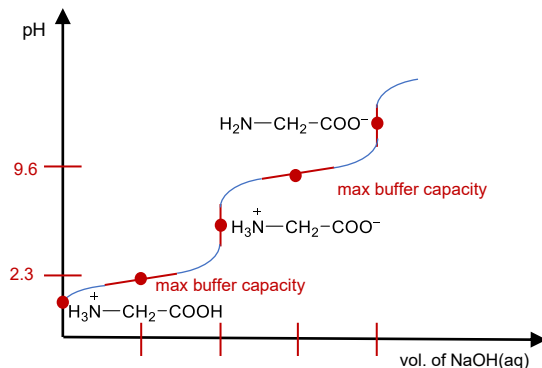
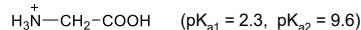
Describe the physical properties of $\text{H}_3\text{N}^+-\text{CH}_2-\text{COO}^-$

melting point	high, due to strong ionic bonds , which require large amount of energy to overcome
solubility in water	high, due to formation of ion-dipole interactions with water molecules

Aqueous amino acids can be neutral or charged, depending on pH



Sketch the titration curve when NaOH(aq) is added to



Relative basicity

Label the following from least to most basic

CH_3NH_2	NH_3	$\text{C}_6\text{H}_5\text{NH}_2$	$\text{CH}_3-\text{C}(=\text{O})-\text{NH}_2$
most basic			least basic

Explanation:

The greater the **availability** of lone pair of electrons on N to coordinate with a proton, the greater the basicity.

CH_3NH_2 - electron-donating alkyl group bonded to N - increases the electron density on N atom - hence lone pair of electrons on N more available for coordination with proton	$\text{C}_6\text{H}_5\text{NH}_2$ - orbital of N atom containing lone pair of electrons overlaps with π electron cloud of benzene ring - lone pair of electrons on N delocalises into benzene ring - hence lone pair of electrons on N less available for coordination with proton	$\text{CH}_3-\text{C}(=\text{O})-\text{NH}_2$ - orbital of N atom containing lone pair of electrons overlaps with π electron cloud of C=O bond - lone pair of electrons on N delocalises into C=O bond - hence lone pair of electrons on N is not available for coordination with proton
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Test

Compound	Reagents / Cond ^a	Observation/ explanation
CH_3NH_2 amine with 5C or less	NaOH(aq), warm/ heat	Observation: pungent gas evolved turned damp red litmus paper blue Explanation: amine is volatile, vapourises upon warming (Note: NaOH is not actually required)
$\text{CH}_3(\text{CH}_2)_5\text{NH}_2$ amine with 6C or more	NaOH(aq), warm/ heat	Observation: no pungent gas evolved Explanation: amine is not volatile, does not vaporise upon warming
$\text{CH}_3-\text{C}(=\text{O})-\text{NH}_2$ Amide	NaOH(aq), <u>heat</u>	Observation: pungent gas evolved turned damp red litmus paper blue Explanation: amide undergo base hydrolysis to form volatile NH_3 which vaporises
$\text{C}_6\text{H}_5-\text{N}(\text{H})-\text{C}(=\text{O})\text{CH}_3$ Amide	NaOH(aq), <u>heat</u>	Observation: no pungent gas evolved Explanation: amide undergo base hydrolysis to form non-volatile phenylamine (has 6 C)
$\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_3^+$ ammonium salt	NaOH(aq), warm/ heat	Observation: pungent gas evolved turned damp red litmus paper blue Explanation: ammonium ion undergo acid-base reaction to form volatile $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ which vaporises